

Quantum Mechanics

Origin of Quantum Mechanics

- Before 1900 Century, most of the phenomenon could be explained by classical physics which is based on Newton's three laws (Classical Mechanics), Thermodynamics and Electromagnetic theory.
- Classical Physics explained satisfactorily the motion of bodies moving with non-relativistic speeds ($v \ll c$). However it is not valid for relativistic speeds ($v \rightarrow c$).
- Classical Physics could not explain following experimentally observed phenomenon.
 - i. Stability of atoms
 - ii. Spectral distribution of black body radiation
 - iii. Specific heat of solids at low temperature
 - iv. Photo-electric effect
 - v. Compton Effect
 - vi. Optical line spectra
- Discovery of electron, proton and neutron motivated scientists to think at atomic level in order to explain these observations.

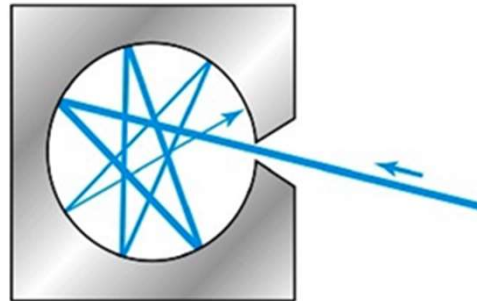
Max Planck (1900) introduced the new concept that the emission and absorption of electromagnetic radiation takes place as discrete quanta, each of which contains a fixed amount of energy. This concept has originated the Quantum mechanics. Using quantum concept Planck explained the black body radiation spectrum. Later Einstein explained the Photoelectric effect in 1905.

Black body

“An ideal body that absorbs all the radiation incidents upon it regardless of wavelength or frequency is called as **black body**.” No radiation passes through it and none is reflected. Such object appears black when it is cold. When the black body is heated, it emits temperature-dependent spectrum of light. This thermal radiation from a black body is termed as black body radiation.

Practical black body

Practically, a black body can be approximated by a hollow object with a very small hole leading to its interior.



Any radiation (irrespective of wavelength) striking the hole enters the cavity, where it is trapped by multiple reflections back and forth until it is absorbed.

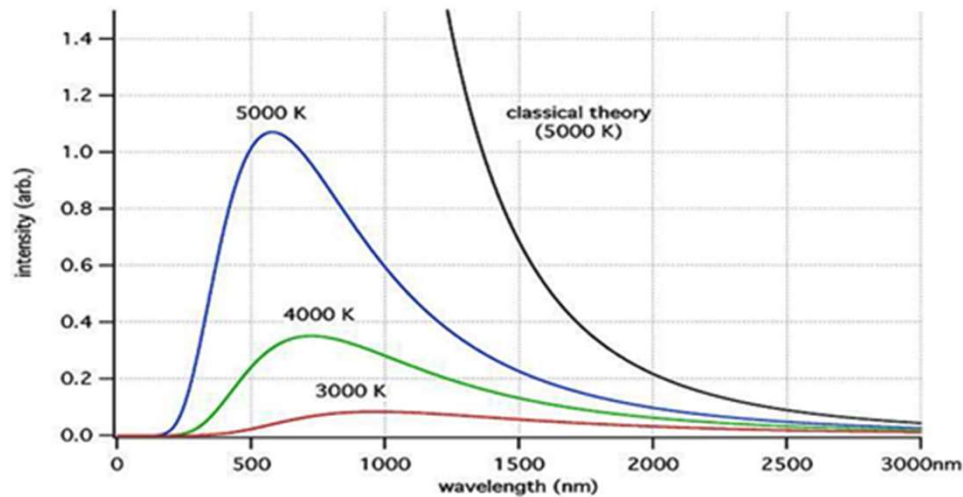
White body

It is a hypothetical object whose surface absorbs no radiation of any wavelength i.e. one which exhibits zero absorption for all wavelengths.

Black Body radiation spectrum

Thermal radiation is similar as light radiation with only difference that its average wavelength is greater than that of the visible light. Spectrum of thermal radiation can be recorded using prism of rock-salt, KCl, etc or a diffraction grating of large spacing.

Lummer and his colleague Pringsheim investigated experimentally the intensity distribution of the emitted radiation by an object at different temperatures. The curve is drawn between intensity and the wavelength of a body heated at different temperatures. The results of the experiment at various temperatures are shown below in the figure.



2.3 Results:

It is observed from the curve that,

- i).** Energy is not uniformly distributed in the Radiation spectrum of radiating source.
- ii).** At a given temperature, the intensity of radiations increases with increase of wavelength and becomes maximum at a particular wavelength. By further increasing the wavelength, intensity of heat radiation decreases.
- iii).** An increase in temperature causes a decrease in $\lambda_{\max}$, where $\lambda_{\max}$ is the wavelength for which the energy emitted is maximum,

$$\lambda_{\max} \propto 1/T$$

$$\lambda_{\max} T = \text{constant} \quad (2.3.1)$$

This is called Wien's Displacement law.

- iv).** The area under each curve represents the total energy emitted by the body at particular temperature for the range of wavelength considered. This area increases with increase of temperature so an increase in temperature causes an increase in energy emission for wavelength.

- v).** It is found that the area under the curve is directly proportional to the fourth power of absolute temperature.

Intensity, I = area under the curve

$$\begin{aligned} I &= \int E_{\lambda} d\lambda \propto T^4 \\ I &= \sigma T^4 \end{aligned} \quad (2.3.2)$$

which is known as Stefan's law.

Planck's Quantum Theory:

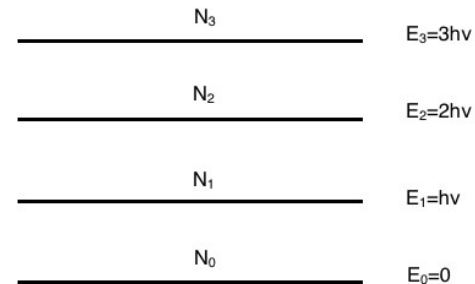
Planck put forward the quantum theory to explain the observed black body radiation spectrum. The assumptions or postulates of his theory are as follows.

1). A blackbody radiation chamber is filled up not only with radiation but also with the simple harmonic oscillators or resonators of the molecular dimensions which cannot have any energy value, but only energy value given by, **$E = nh\nu$, $n=0,1,2,3,-----$**

where ν is the frequency of oscillator, h is plank's constant (6.63×10^{-34} J-Sec)

2). The oscillators cannot radiate or absorb energy continuously but an oscillator of frequency ν can only radiate or absorb energy in units or quanta of magnitude $h\nu$.

Energy levels of Planck's oscillator



Planck Radiation formula

Plank found an empirical formula to explain the experimentally observed distribution of energy in the spectrum of a blackbody. The formula may be deduced using the above two postulates.

From the statistical mechanics, the number of resonator per unit volume in the frequency range ν and $\nu + d\nu$ is given by

$$N = \frac{8\pi\nu^2}{c^3} d\nu \quad (3.2.1)$$

The average energy of an oscillator is given by

$$\bar{E} = \frac{h\nu}{e^{h\nu/kT} - 1} \quad (3.2.2) \text{ (see section 3.3 for$$

details)

with $E = h\nu$.

The energy density belonging to range $d\nu$ can be obtained by multiplying the average energy of a plank's oscillator by the number of resonators per unit volume in the frequency range ν and $\nu + d\nu$,

$$E_\nu d\nu = \frac{8\pi\nu^2}{c^3} \times \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} d\nu$$

$$E_\nu d\nu = \frac{8\pi\nu^3}{c^3} \frac{h}{e^{\frac{h\nu}{kT}} - 1} d\nu \quad (3.2.3)$$

$E_\nu d\nu$ is the energy density, which is the total energy per unit volume belonging to range $d\nu$. This equation gives the plank's radiation law.

In terms of wavelength, energy density $E_\lambda d\lambda$ belonging to range λ and $\lambda+d\lambda$ will be

$$E_\lambda d\lambda = \frac{8\pi h}{c^3} \left(\frac{c}{\lambda}\right)^3 \frac{1}{e^{\frac{h\nu}{kT}} - 1} \cdot \frac{c}{\lambda^2} d\lambda$$

$$\text{Because, } \nu = \frac{c}{\lambda} \quad |d\nu| = \left| -\frac{c}{\lambda^2} d\lambda \right| = \left| \frac{c}{\lambda^2} d\lambda \right|$$

$$E_{\lambda} d\lambda = \frac{8\pi h c}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1} d\lambda \quad \text{----- (3.2.4)}$$

which gives the energy density for wavelength in range λ and $\lambda+d\lambda$ in the spectrum of black body. This formula fits well with the experimentally observed black body radiation spectrum and explains its features.

Derivation of Average energy of a Planck's oscillator:

According to second postulates of Plank's the exchange of energy between radiation and matter cannot take place continuously, but are limited to discrete set of values $0, h\nu, 2h\nu, \dots, nh\nu$ etc, so that the energy content of one mode is $nh\nu$, where n is an integer number $n=0,1,2,3, \dots$

The number of oscillators N_n having the energy $nh\nu$ is given by Boltzmann distribution law,

$$N_n = N_0 e^{\frac{-nh\nu}{kT}} \quad \text{----- (3.3.1)}$$

The energy contributed by N_n oscillators,

$$nh\nu N_n = nh\nu N_0 e^{\frac{-nh\nu}{kT}} \quad \text{----- (3.3.2)}$$

Therefore the average energy of an oscillator is,

$$\bar{E} = \frac{\sum_{n=0}^{\infty} nh\nu N_0 e^{\frac{-nh\nu}{kT}}}{\sum_{n=0}^{\infty} N_0 e^{\frac{-nh\nu}{kT}}} \quad \text{---- (3.3.3)}$$

Since n is an integer,

$$\bar{E} = \frac{0 + h\nu e^{\frac{-h\nu}{kT}} + 2h\nu e^{\frac{-2h\nu}{kT}} + \dots}{1 + e^{\frac{-h\nu}{kT}} + e^{\frac{-2h\nu}{kT}} + \dots} \quad \text{----- (3.3.4)}$$

Let $x = e^{\frac{-h\nu}{kT}}$ then equation (4) can be written as,

$$\bar{E} = h\nu x \frac{(1 + 2x + 3x^2 + \dots \infty)}{1 + x + x^2 + \dots \infty}$$

or

$$\bar{E} = h\nu x \frac{(1 - x)^2}{(1 - x)^{-1}}$$

$$\text{i.e. } \bar{E} = \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} \quad \text{Because } x = e^{\frac{-h\nu}{kT}}$$

This is the average energy of an oscillator of frequency ν in thermal equilibrium at temperature T.

Compton Effect

(A. H. Compton, 1923)

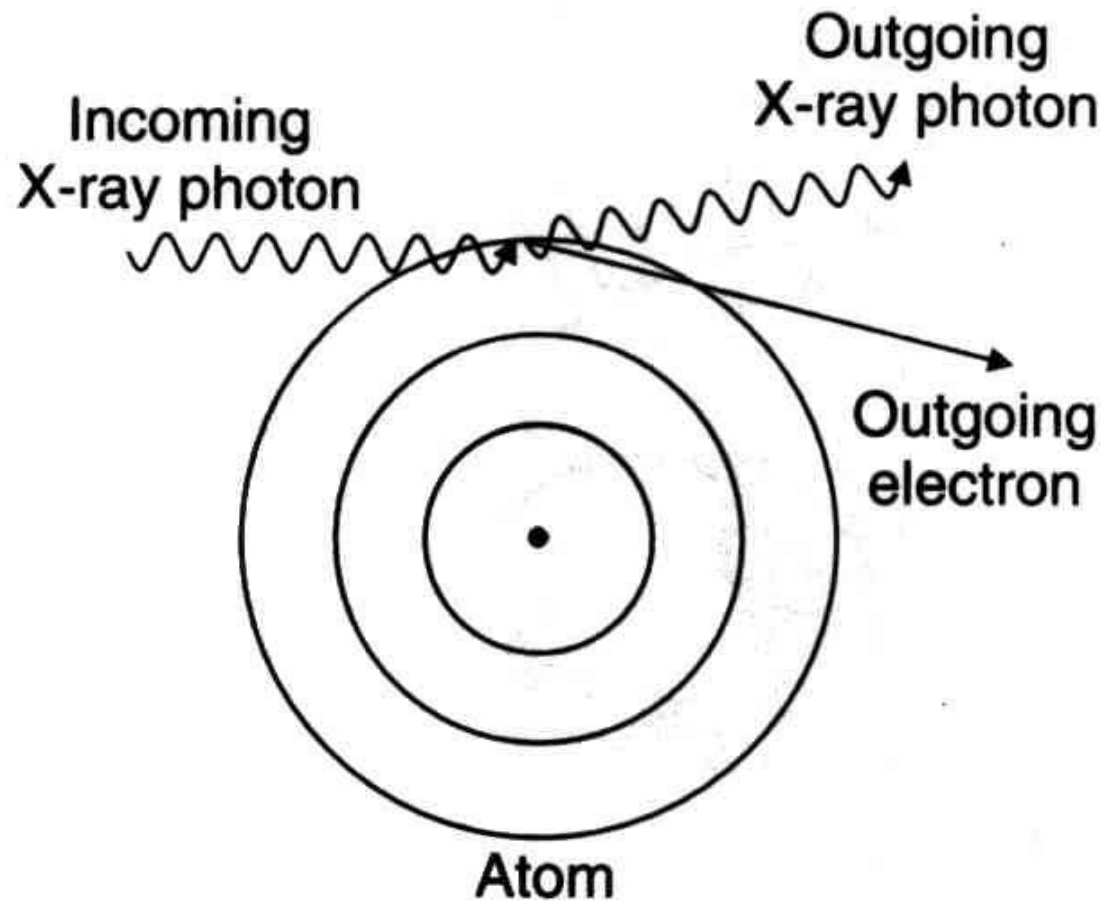
- The Compton effect was first demonstrated in 1923 by [Arthur Holly Compton](#) (for which he received a 1927 Nobel Prize in Physics). Compton's graduate student, Y.H. Woo, later verified the effect.
- This is an experiment to demonstrate interaction of radiation and matter.
- Compton effect verifies the particle nature of light. Light travels in the form of photons ($E=h\nu$): massless particles.

Energy of incoming X-ray photon $\cong$ 17000 eV, that can be transferred to electron on collision.

Binding (orbital or ionization) energy of electron in outermost shell $\cong$ 10-15 eV

So the electron is considered as free electron.

For Compton scattering, free electron at rest is considered. Generally, **a metal target** having free electrons or loosely bound electrons is considered to observe Compton Scattering.



What is Compton Effect or Compton Scattering?

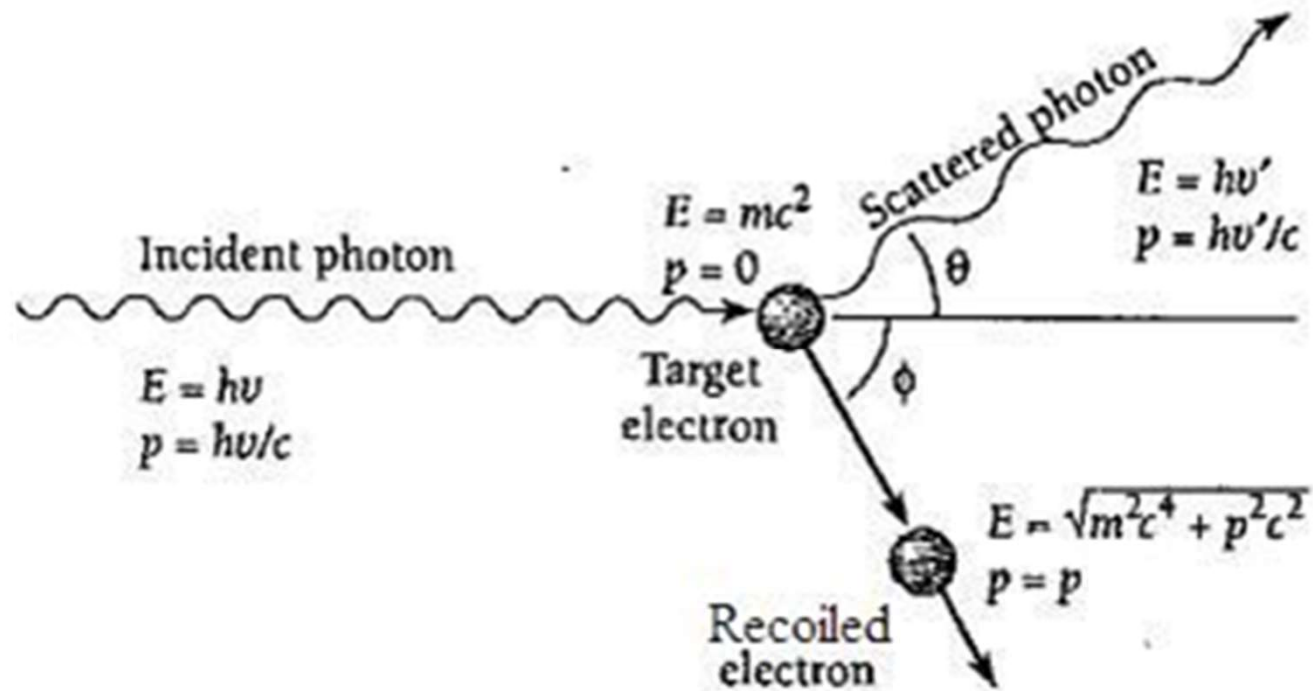
When a beam of high frequency monochromatic radiation (e.g. X-rays, γ -rays) is scattered by a target electron of light element such as Carbon, the scattered radiations contain radiation of lower frequencies together with small extent of unchanged radiation.

The target electron is recoiled in some other direction.

Compton observed that the difference in the frequencies (or wavelengths) of these scattered radiation increases with the angle of scattering. It is independent of the incident beam frequency and also the nature of scattering material.

This phenomenon is called the 'Compton Effect'. It has been explained satisfactorily on the basis of quantum theory.

Experimental Arrangement



Correct $m = m_0$ in the figure.

Derivation of Compton shift

We know that the momentum and energy both remain conserved in this collision process. Conservation of momentum along and perpendicular to the direction of incidence will be,

$$\frac{h\nu}{c} = \frac{h\nu'}{c} \cos \theta + p \cos \phi \quad (4.1)$$

$$0 = \frac{h\nu'}{c} \sin \theta - p \sin \phi \quad (4.2)$$

Conservation of energy yields,

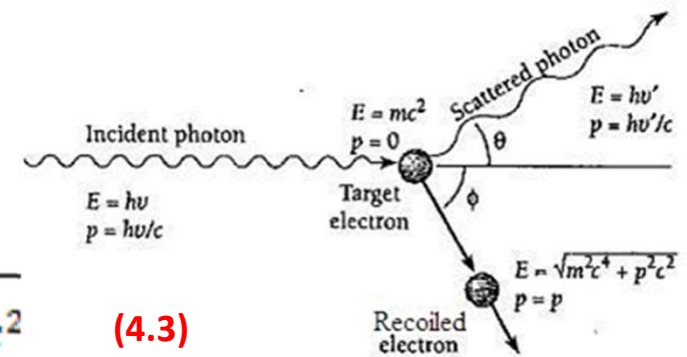
$$h\nu + m_0 c^2 = h\nu' + \sqrt{p^2 c^2 + m_0^2 c^4} \quad (4.3)$$

$$h(\nu - \nu') + m_0 c^2 = \sqrt{p^2 c^2 + m_0^2 c^4}$$

Squaring both side,

$$h^2(\nu - \nu')^2 + m_0^2 c^4 + 2m_0 c^2 h(\nu - \nu') = p^2 c^2 + m_0^2 c^4$$

$$h^2(\nu - \nu')^2 + 2m_0 c^2 h(\nu - \nu') = p^2 c^2 \quad (A)$$



Now from equation (4.1) and (4.2)

$$p \cos \varphi = \frac{h(\nu - \nu' \cos \theta)}{c} \quad \text{----- (4.4)}$$

and,

$$p \sin \varphi = \frac{h\nu'}{c} \sin \theta \quad \text{----- (4.5)}$$

After squaring and adding equation (4.4) and (4.5) we will get,

$$p^2 = \left(\frac{h}{c}\right)^2 [(\nu - \nu' \cos \theta)^2 + \nu'^2 \sin^2 \theta]$$

$$p^2 = \left(\frac{h}{c}\right)^2 [\nu^2 + \nu'^2 \cos^2 \theta - 2\nu\nu' \cos \theta + \nu'^2 \sin^2 \theta]$$

$$p^2 c^2 = h^2 [\nu^2 + \nu'^2 - 2\nu\nu' \cos \theta] \quad \text{----- (4.6) (B)}$$

$$h^2 [\nu^2 + \nu'^2 - 2\nu\nu'\cos\theta] = h^2(\nu - \nu')^2 + 2m_0 c^2 h(\nu - \nu')$$

$$h\nu\nu'(1 - \cos\theta) = m_0 c^2 h(\nu - \nu')$$

$$\frac{\nu - \nu'}{\nu\nu'} = \frac{h}{c^2 m_0} (1 - \cos\theta)$$

$$\frac{1}{\nu'} - \frac{1}{\nu} = \frac{h}{c^2 m_0} (1 - \cos\theta)$$

$$\frac{c}{\nu'} - \frac{c}{\nu} = \frac{h}{cm_0} (1 - \cos\theta)$$

$$\lambda' - \lambda = \frac{h}{cm_0} (1 - \cos\theta) \quad \text{----- (4.7)}$$

where $\frac{h}{cm_0}$ is called the Compton Wavelength λ_c .

$\lambda' - \lambda = \Delta\lambda$, is known as Compton shift

$$\text{Compton wavelength, } \lambda_c = \frac{h}{cm_0} \quad \text{----- (4.8)}$$

Equation (7) gives the change in wavelength expected for a photon that is scattered through the angle θ by an electron of rest mass m_0 . This change is independent of the wavelength λ of the incident photon. Equation (7) can also be written as,

$$\lambda' = \lambda + \lambda_c (1 - \cos \theta) \quad \text{----- (4.9)}$$

Compton wavelength for electron is

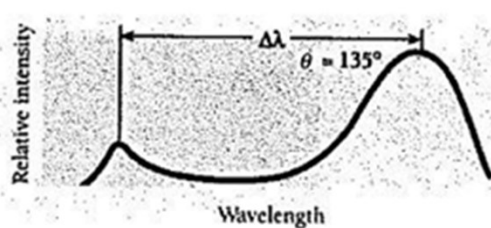
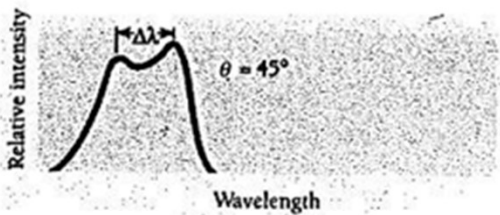
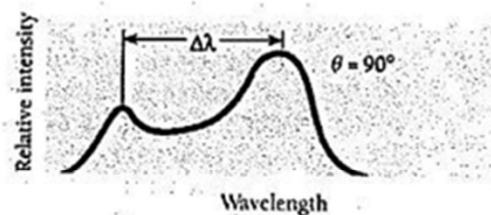
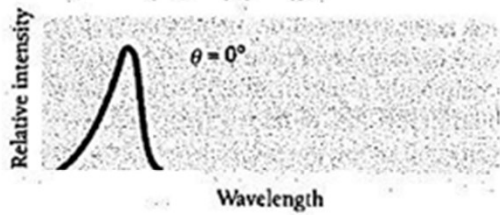
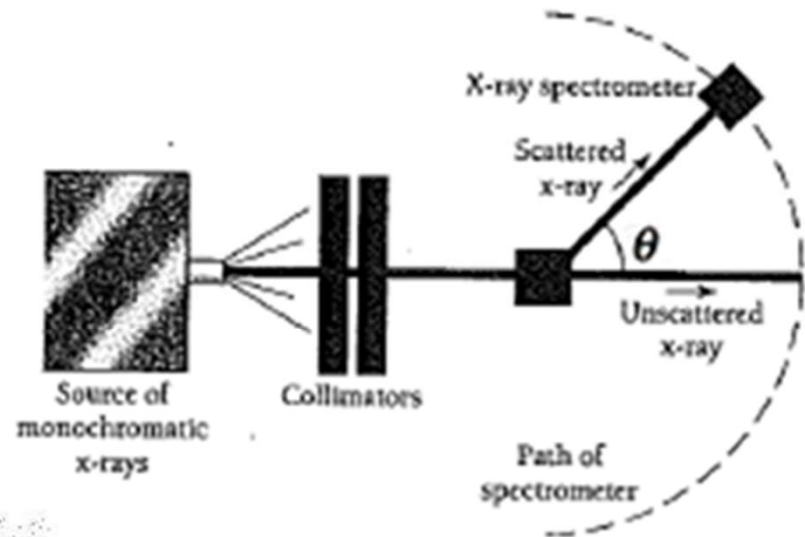
$$\lambda_c = 2.426 \times 10^{-12} \text{ m}$$

The change in the wavelength ranges from 0 at $\theta=0^\circ$ to twice the Compton wavelength at $\theta= 180^\circ$. (Maximum Compton shift = $2 \times 2.426 \times 10^{-12} \text{ m} = 0.048 \text{ \AA}$)

The change of wavelength is appreciably observed only in X-ray. The shift in wavelength for visible light is less than 0.01% of the initial wavelength, whereas for X-ray of $\lambda = 0.1 \text{ nm}$, it is several percent. So Compton effect observed only with low wavelength or high frequency radiations.

Experimental verification of Compton effect:

In this experimental arrangement, monochromatic X-ray (i.e. X-ray of single known λ) are collimated by two lead blocks with holes and then are directed at a target (i.e. scattering substance, Graphite crystal). To record the intensity of scattered X-ray, an X-ray spectrometer (or detector) moves in the arc of circle with centre at the target.



The detector is used to determine the wavelength of the scattered ray at different angle θ . The results of X-ray diffraction experiment are shown. The first peak shows unmodified radiation while second peak shows the modified radiation.

Reason for existence of unmodified radiation

The electrons when hit the target, modify the photon wavelength which also require a change of energy. So such electron must be either free or loosely bound. If a photon encounters a tightly bound electron, the whole atom is involved in the collision process and the value of m_0 in $\lambda' - \lambda = \frac{h}{cm_0}(1 - \cos\theta)$ becomes the mass of atom rather than the mass of electron. Since the mass of atom is several thousand times the electric mass, the Compton wavelength shift $\lambda' - \lambda$ will be too small to detect. This is the source of observed unmodified radiation after collision.

Wave-Particle Duality

- Wave Nature of Light

Interference, Diffraction, Polarization

(Wave Properties: Wavelength, Frequency, Amplitude, Energy)

- Particle Nature of Light

Photo-electric effect, Compton Effect

(Particle Properties: Position, Mass, Energy, Momentum, Kinetic energy, Electric charge)

It is known as dual nature (Wave & Particle) of light or any EM radiation.

In 1924, **Louis de Broglie** suggested the matter also exhibits dual nature as well similar as light. He said that material particle like electron might act at certain times like waves. The waves associated with the material particles are known as **matter waves** or **de Broglie waves**.

For a photon $E = h\nu$ and $p = \frac{h\nu}{c} = \frac{h}{\lambda}$

$$\therefore \lambda = \frac{h}{p}$$

A material particle will have associated de Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where m is the mass and v is the velocity of the particle. De Broglie concluded that a moving body behaves in certain ways as it has a wave nature.

The above relation may also be written in terms of kinetic energy K ,

We know that,

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$\therefore p = \sqrt{2mK}$$

$$\therefore \lambda = \frac{h}{\sqrt{2mK}}$$

When a charged particle carrying a charge q is accelerated through a potential difference of V volts, then kinetic energy K is given by,

$$K = qV$$

Hence the de Broglie wavelength will be,

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

If the material particles are in thermal equilibrium at temperature T , then

$$K = \frac{3}{2}kT$$

where $k = 1.38 \times 10^{-23} \text{ J/}^\circ\text{K}$, is the Boltmann constant.

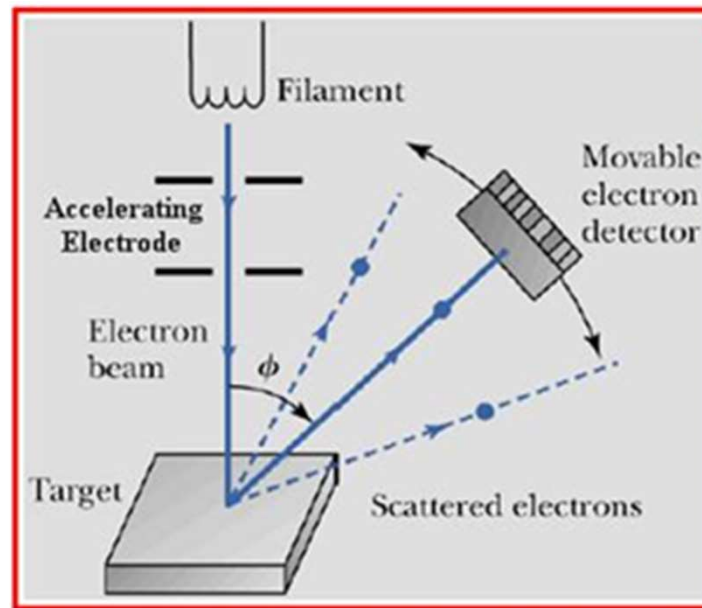
and de Broglie wavelength will be,

$$\lambda = \frac{h}{\sqrt{3mkT}}$$

Experimental verification of de Broglie theory of matter waves

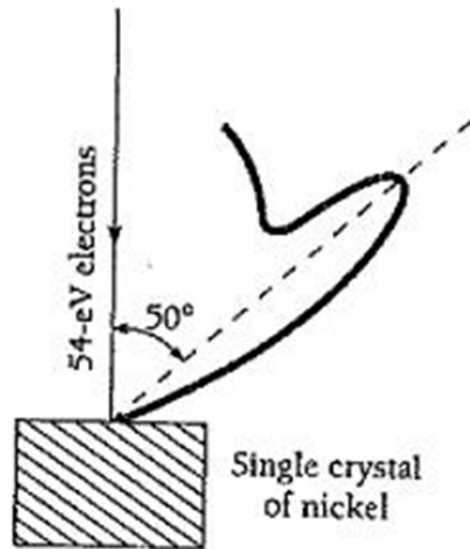
Devisson and Germer Experiment (1927)

They demonstrated that the electron beam undergo diffraction in the regular atomic arrays of Ni crystal.



θ is the angle between incident beam and crystal plane
 ϕ is the angle between incident beam and diffracted beam

De Broglie Wavelength associated with the electron beam



$$\lambda = \frac{h}{\sqrt{2mqV}}$$

putting for electron, $q = 1.6 \times 10^{-19}$ coulomb

$V = 54$ volts

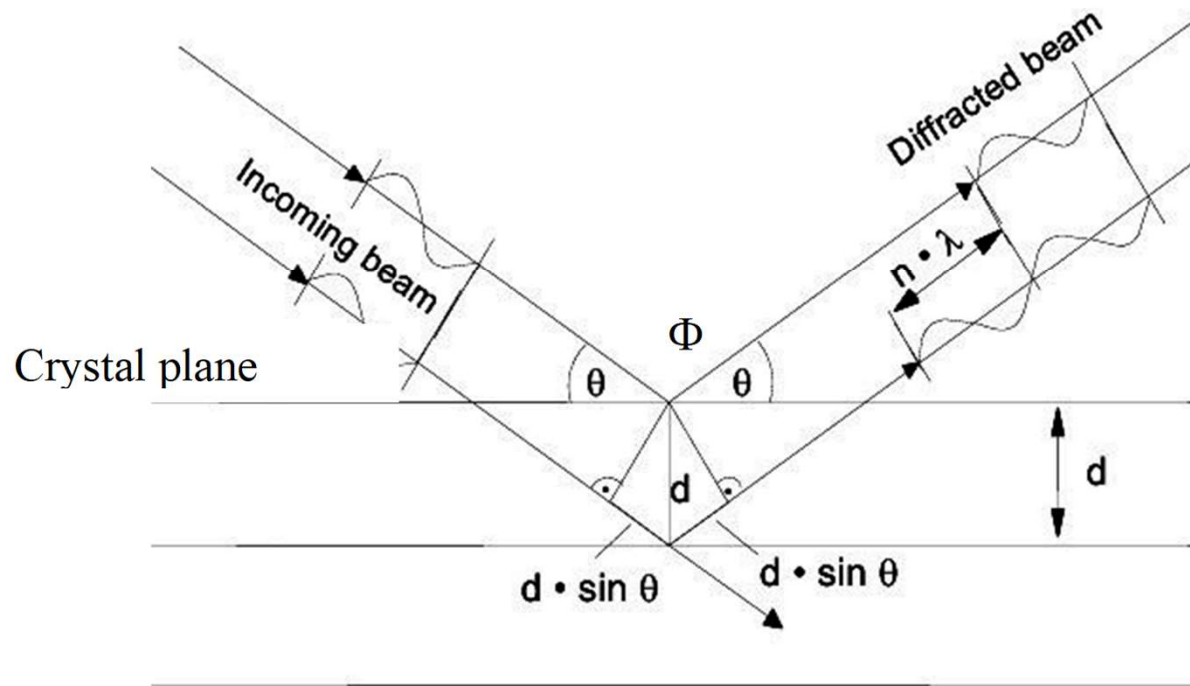
$m = 1.9 \times 10^{-31}$ Kg

Estimated, $\lambda = 0.166$ nm

The pattern of the observed spectrum

Analysis of Crystal diffraction

From Davisson and Germer experiment, the angles of incidence and scattering relative to the family of Bragg plane is shown below.



The Bragg law for X-ray diffraction by a crystal is

$$2d \sin \theta = n\lambda$$

where d is the spacing between planes in the crystal, λ is the wavelength of X-rays and θ is the angle at which maximum diffraction occurs.

$$\text{At } \Phi = 50^\circ, \text{ angle } \theta = (180 - \Phi)/2 = 65^\circ$$

The spacing of the planes which can be measured by X-ray diffraction is 0.091 nm. We can now use Bragg equation for maxima in the diffraction pattern as,

$$n\lambda = 2d \sin \theta$$

Substituting, $d=0.091\text{nm}$,
 $\theta = 65^\circ$
and $n=1$ (first order)

$$\begin{aligned} \text{we can get, } \lambda &= 2 \times 0.091 \sin 65^\circ \\ \lambda &= 0.165 \text{ nm} \end{aligned}$$

This is the nearly the same value that is obtained by de-Broglie calculation. Hence this experiment directly verifies de Broglie hypothesis or the wave nature of moving bodies.

Wave function and its importance

- Wave function (ψ) represents the properties of particle in Quantum mechanics.
- The properties include position, energy, momentum etc.
- ψ itself has no physical interpretation. However, $|\psi|^2$ is proportional to the probability of finding the particle there at that time. (Max Born hypothesis)

$$p \propto |\psi|^2$$

- Wave functions are usually complex with both real and imaginary parts. However, the probability is a positive real quantity.
- The probability density, $pd = \psi \cdot \psi^*$

ψ can be written in the form

$$\psi = A + iB$$

Where A and B are the real function. The complex conjugate ψ^* of ψ is

$$\psi^* = A - iB$$

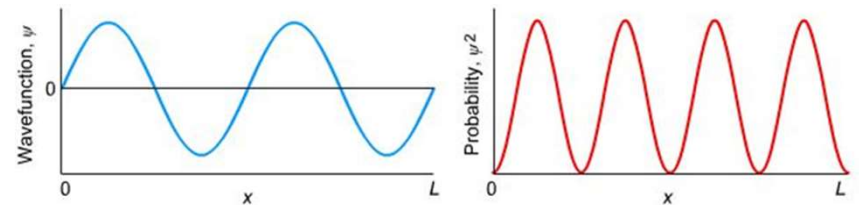
Therefore,

$$|\psi|^2 = \psi^* \psi = A^2 - i^2 B^2, \text{ but } i^2 = -1$$

Or

$$|\psi|^2 = A^2 + B^2$$

Hence $|\psi|^2 = \psi^* \psi$ is always a positive real quantity.



Probability = 0, Particle is absent

Probability = 1, Particle is certainly present

Normalized wave function

Probability of finding the particle in somewhere in space at particular time is obtained by integration of probability density under the desired limit.

$$P = \int_{-\infty}^{\infty} |\psi|^2 dv$$

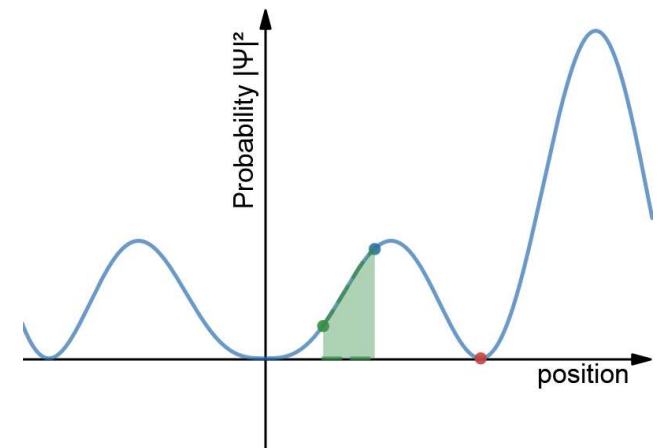
where dv is small volume element given by $dv = dx dy dz$

$$\int_{-\infty}^{\infty} |\psi|^2 dv = 0$$

Particle is not present.

$$\int_{-\infty}^{\infty} |\psi|^2 dv = 1$$

Particle is certainly present.



Wave functions that follow this condition (certain probability) are known as normalized wave functions.

Ex. If we put a particle in a box, then it is certainly present within the interior of box. The wave function that will represent the particle in the box will be normalized.

Well-behaved wave function

The wave functions ψ having the following properties are admissible as mathematical reorientations of real bodies.

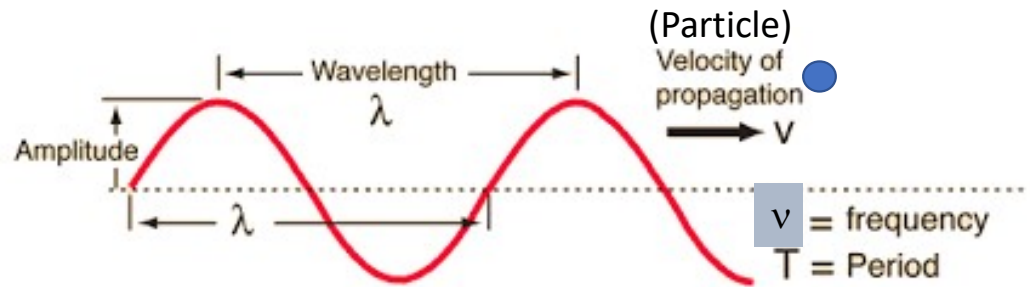
1. ψ must be continuous and single valued.
2. $\frac{\partial \psi}{\partial x}, \frac{\partial \psi}{\partial y}, \frac{\partial \psi}{\partial z}$ must be continuous and single valued everywhere.
3. ψ must be normalized, which means ψ must go to 0 as $x \rightarrow \pm\infty, y \rightarrow \pm\infty, z \rightarrow \pm\infty$ in order that $\int |\psi|^2 dv$ over all space to be a finite constant.

A well-behaved normalized wave function of a specific system determines the probability distribution function for that function. This probability density is not directly observed experimentally for a single particle. So the well-behaved function is the mathematical representation of a physical system.

Wave and Particle velocity

If u is the velocity of wave than

$$u = v\lambda$$



Where v is the frequency and λ is the de Broglie wavelength

$$\lambda = \frac{h}{mv}$$

where v is the particle velocity.

The frequency v is found from the relation

$$E = hv = mc^2$$

$$v = \frac{mc^2}{h}$$

Now velocity of de Broglie wave,

$$u = v\lambda = \frac{mc^2}{h} \lambda$$
$$\therefore u = \frac{c^2}{v}$$

Now since $v < c$, above relation implies that $u > c$ or $u > v$.

Thus the velocity of de Broglie wave comes larger than the velocity of particle itself. In this case the particle will be left behind the associated wave.

Phase velocity:

The usual form of the wave equation in general is

$$y = A \cos(\omega t - kx)$$

where y is the displacement at time t and place x

ω is the angular frequency $= 2\pi\nu$

k is the wave number $= 2\pi/\lambda$ (radiation/m)

The phase velocity of a wave is given by,

$$v_p = \frac{\omega}{k}$$

This is the velocity with which the planes of constant phase move.

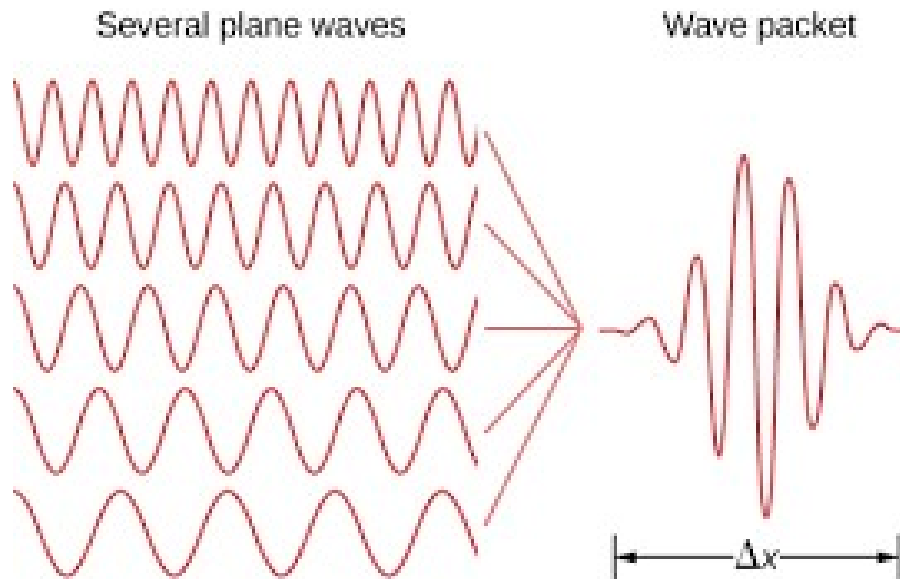
Phase = constant

$\omega t - kx = \text{constant}$

$\omega \cdot dt - k \cdot dx = 0$

$dx/dt = v_p = \omega/k$

Particle in a wave packet



The exact position of particle within the wave packet is not known.



$$G = \frac{d\omega}{dk} = \frac{\omega_1 - \omega_2}{k_1 - k_2} \quad (\text{Group velocity})$$

Heisenberg Uncertainty principle

Heisenberg uncertainty principle

It is impossible to determine the exact position and momentum of a particle simultaneously.



Werner Heisenberg,
1927

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

Δx = Uncertainty of Position

Δp = Uncertainty of Momentum

$$\therefore \lambda = \frac{h}{p}$$



Figure 1.2a : Narrow wave Packet

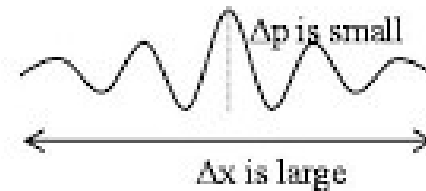


Figure 1.2b : Wide wave

$$\text{taking } \hbar = \frac{h}{2\pi}$$

$$\Delta x \Delta p_x \geq \frac{\hbar}{2}$$

Similarly we can also write the y- component of uncertainty as

$$\Delta y \cdot \Delta p_y \geq \frac{\hbar}{2} \quad \text{----- (8.1.6)}$$

z- Component of uncertainty as

$$\Delta z \cdot \Delta p_z \geq \frac{\hbar}{2} \quad \text{----- (8.1.7)}$$

It is to be noted that the uncertainty principle will be valid when both the position and momentum is measured in the same direction either x or y or z.

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi} \quad \text{Energy-Time relation}$$

Any two physical quantities having product of units as “Joule.sec” will follow the Heisenberg uncertainty principle.

Wave group:

The difficulty of the outcome $u > c$ or $u > v$, can be overcome by assuming each moving particle of matter consists of a group of waves or a wave packet, rather than a single wave. A wave group of certain wavelength consists of a number of component waves of slightly different wavelengths superimposed upon each other. The mutual interference of waves leads to amplitude variation and defines the shape of wave packet as shown below.

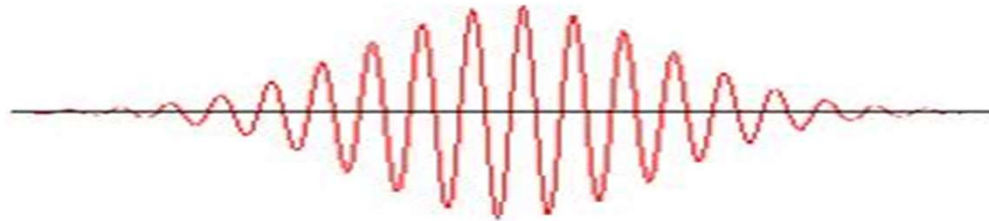


Fig. 9. Schematic of the wave group

This group of wave travels in the medium with group velocity G . Considering the superposition of two wave components, the group velocity is given by,

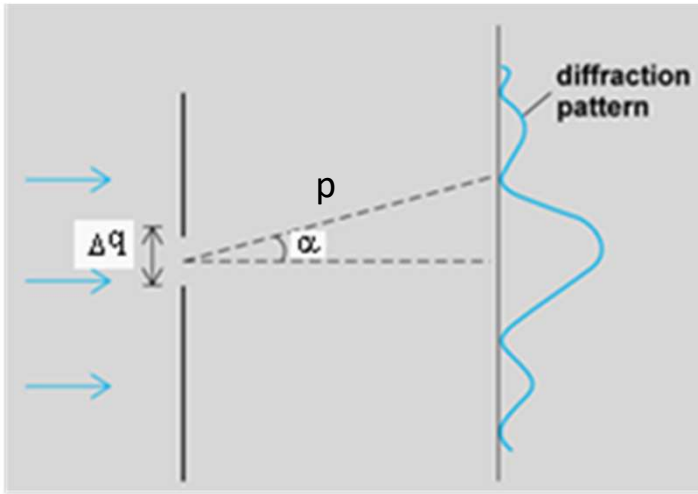
$$G = \frac{d\omega}{dk} = \frac{\omega_1 - \omega_2}{k_1 - k_2}$$

It can also be easily found that the group velocity is same as the particle velocity.

$$G = v$$

Hence the position of the moving particle can be anywhere inside this wave packet.

Application of H. U. P. in Single Slit diffraction experiment



A beam of particles (electrons) of momentum p is incident on the screen after diffraction through a slit of width Δq .

The particles are associated with matter waves, each of wavelength h/p .

The width of the slit Δq determines the degree of precision with which the position of electron is known along y-direction.

Diffraction occurs at the slit and a pattern corresponding to a rectangular aperture is obtained.

The uncertainty in the momentum Δp can be obtained by y-component of the momentum of diffracted electron beam.

As angle α varies from $+\alpha$ to $-\alpha$,
 $\Delta p = p \sin \alpha - (-p \sin \alpha) = 2p \sin \alpha$

The angular width of the central maxima ($n=1$) is given by,

$$\sin \alpha = \frac{\lambda}{\Delta q}$$

$$\begin{aligned} \Delta q \cdot \Delta p &= \Delta q \cdot 2p \sin \alpha \\ &= \Delta q \cdot 2p \cdot \lambda / \Delta q = 2p \cdot \lambda \\ &= 2p \cdot (h/p) \quad (\text{de-Broglie wavelength}) \\ &= 2h \end{aligned}$$

$$\Delta q \cdot \Delta p \geq h/4\pi$$

Why electron can not reside inside the nucleus?

Another application of Heisenberg uncertainty principle.
(Refer tutorial sheet/book for solution).

Expectation value

Considering a group of many particles which are distributed randomly along x-direction. N1 particles are at x1, N2 particles are at x2, N3 are at x3 and so on.

Then the average position is

$$\bar{x} = \frac{N_1 x_1 + N_2 x_2 + N_3 x_3 + \dots}{N_1 + N_2 + N_3 + \dots} = \frac{\sum N_i x_i}{\sum N_i}$$

Now, suppose we are having only one particle present in the similar x-space, then it is advisable to replace numbers N1, N2, N3, by probabilities of finding that particles at different locations x1, x2, x3,

This probability is $P_i = |\psi_i|^2 dx$ where, ψ_i is particle wave evaluated at $x = x_i$.

In this case, the average value is replaced by Expectation value.

Therefore, we can define the expectation value of position $\langle x \rangle$ as

$$\langle x \rangle = \frac{\int_{-\infty}^{\infty} x |\psi|^2 dx}{\int_{-\infty}^{\infty} |\psi|^2 dx}$$

or $\langle x \rangle = \frac{\int_{-\infty}^{\infty} \psi^* x \psi dx}{\int_{-\infty}^{\infty} |\psi|^2 dx}$

If ψ is normalized wave function, then

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 1$$

Therefore,

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi^* x \psi dx$$

Expectation value of an arbitrary function $f(x)$ will be,

$$\langle f(x) \rangle = \int_{-\infty}^{\infty} \psi^* f(x) \psi dx$$

Expectation values for momentum and energy:

According to the general convention,

Expectation value for momentum p should be

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi^* p \psi dx$$

and expectation value for energy E should be

$$\langle E \rangle = \int_{-\infty}^{\infty} \psi^* E \psi dx$$

However, these relations are not valid to represent expectation values for momentum and energy due to uncertainty principle.

If we specify position x , i.e. $\Delta x=0$, hence we can not specify momentum p simultaneously with uncertainty principle ($\Delta x \Delta p_x \geq \frac{\hbar}{2}$). That means no function like $p(x)$ exists.

It is similar for energy because of $\Delta E \Delta t \geq \frac{h}{4\pi}$. If we specify time t , the function $E(t)$ is impossible.

So how to get momentum and energy expectation values?

(Operator method)

We can describe a moving particle in terms of the probability amplitude ψ which may be called as wave function. From the wave mechanics one can assume that ψ for a particle moving freely in the +x direction is specified by,

$$\begin{aligned}\psi &= A \exp[i(kx - \omega t)] \\ \psi &= A \exp\left[2\pi i\left(\frac{x}{\lambda} - \nu t\right)\right]\end{aligned}\tag{9.6.1}$$

But we know that,

$$\begin{aligned}E &= h\nu = 2\pi\eta\nu & \text{or } \nu &= \frac{E}{2\pi\eta} \\ \text{and } \lambda &= \frac{h}{p} = \frac{2\pi\eta}{p} & \text{or } \frac{1}{\lambda} &= \frac{p}{2\pi\eta}\end{aligned}$$

Replace η by $\hbar$ ($h/2\pi$).

Hence from equation (9.6.1),

$$\begin{aligned}\psi &= A \exp\left[\frac{2\pi i}{2\pi\eta}(xp - Et)\right] \\ \psi &= A \exp\left[\frac{i}{\eta}(xp - Et)\right]\end{aligned}\tag{9.6.2}$$

$$\begin{aligned}
\frac{\partial \psi}{\partial x} &= \frac{i}{\eta} p \psi \\
\text{or} \quad \frac{\partial}{\partial x} &= \frac{i}{\eta} p \\
\text{or} \quad p &= -\frac{i}{\eta} \frac{\partial}{\partial x} \quad (9.6.3)
\end{aligned}$$

After differentiating equation (9.6.2) with respect to t, we will get,

$$\begin{aligned}
\frac{\partial \psi}{\partial t} &= -\frac{i}{\eta} E \psi \\
\Rightarrow \frac{\partial}{\partial t} &= -\frac{i}{\eta} E \quad \text{or} \quad E = -\frac{\eta}{i} \frac{\partial}{\partial t} = i\eta \frac{\partial}{\partial t} \\
E &= i\eta \frac{\partial}{\partial t} \quad (9.6.4)
\end{aligned}$$

The energy E and momentum p are called **observables** are now replaced by **operators** which operate on the wave function. So in non-relativistic quantum mechanics the momentum is replaced by the momentum operator,

$$\hat{p} = \frac{\eta}{i} \frac{\partial}{\partial x} \quad (9.6.5)$$

And the energy is replaced by the energy operator,

$$\hat{E} = i\eta \frac{\partial}{\partial t} \quad (9.6.6)$$

We can use these operators to obtain expectation values of p and E . These are given by,

$$\langle p \rangle = \int_{-\infty}^{\infty} \psi^* \hat{p} \psi \, dx$$

$$\langle E \rangle = \int_{-\infty}^{\infty} \psi^* \hat{E} \psi \, dx$$

In general, the expectation value of an operator is given by,

$$\langle G(x, p) \rangle = \int_{-\infty}^{\infty} \psi^* \hat{G} \psi \, dx$$

In this way all the information about a system that is permitted by the uncertainty principle can be obtained from its wave function Ψ .

Schrödinger wave equation



Erwin Schrödinger, 1926
Nobel Prize in Physics-1933

In 1926 Erwin **Schrödinger**, an Austrian physicist, took the Bohr atom model one step further. **Schrödinger** used mathematical equations to describe the likelihood of finding an electron in a certain position. This atomic model is known as the quantum mechanical model of the atom.

Based on de Broglie's idea that particles could exhibit wavelike behavior, Austrian physicist Erwin **Schrödinger** theorized that the behavior of **electrons** within atoms could be explained by treating them mathematically as matter waves.

The wave function $\psi(x, y, z, t)$, is dependent on space and time coordinates, represents the amplitude of the matter wave. The equation of matter wave that contains both space and time variation of wave function ψ is known as Schrodinger wave equation. It is represented into two forms.

1. Time dependent wave equation
2. Time independent wave equation

Time dependent Schrodinger wave equation

The total energy of the particle is

$$E = KE + PE$$

$$E = \frac{p^2}{2m} + V(x, y, z, t)$$

where V is the potential energy function. In quantum mechanics energy becomes an operator and hence the total energy operator is called as Hamiltonian of the particle and is represented by H. The Hamiltonian operator H could be obtained from

$$H = \frac{p^2}{2m} + V(x, y, z, t) \text{ (using momentum operator } p \text{ \& } \frac{\partial}{\partial x} = \nabla \text{)}$$

$$H = \frac{(-i\hbar\nabla)^2}{2m} + V$$

$$H = -\frac{\hbar^2}{2m}\nabla^2 + V \quad (10.1)$$

Now H and E are the operators operating on ψ . Hence we can write,

$$E\psi = H\psi \quad (10.2)$$

With operators on the wave function, the equation (10.2) will be,

$$\begin{aligned}i\hbar \frac{\partial \psi}{\partial t} &= -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \\ -\frac{2m}{\hbar^2} i\hbar \frac{\partial \psi}{\partial t} &= \nabla^2 \psi - \frac{2m}{\hbar^2} V\psi \\ \nabla^2 \psi + \frac{2mi}{\hbar} \frac{\partial \psi}{\partial t} - \frac{2m}{\hbar^2} V\psi &= 0\end{aligned}\tag{10.3}$$

This is called *time dependent Schrödinger wave equation*.

Time independent Schrodinger wave equation

If a moving particle has a wave associated with it, then its nature in space and time should be represented by a definite wave equation. If the position coordinates of particle be (x, y, z) and ψ is the periodic displacement for the matter waves at any instant t. The differential equation of motion of the wave can be written as,

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0 \quad (10.1.1.1)$$

where v is the wave velocity.

Solution of equation (10.1.1.1) can be given by,

$$\psi(x, y, z, t) = \psi_0(x, y, z) e^{-i\omega t} \quad (10.1.1.2)$$

where ψ_0 is the amplitude of the particle wave at the point (x, y, z) independent of time.

On differentiating equation (10.1.1.2) twice with respect to time, we will get,

$$\begin{aligned} \frac{\partial^2 \psi}{\partial t^2} &= -\omega^2 \psi_0 e^{-i\omega t} \\ \text{or,} & \\ \frac{\partial^2 \psi}{\partial t^2} &= -\omega^2 \psi \end{aligned} \quad (10.1.1.3)$$

Substituting equation (10.1.1.3) into (10.1.1.1) to get,

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{\omega^2}{v^2} \psi = 0 \quad (10.1.1.4)$$

where,

$$\omega = 2\pi\nu = 2\pi \frac{v}{\lambda}$$

$$\therefore \frac{\omega}{v} = \frac{2\pi}{\lambda} \quad (10.1.1.5)$$

Substituting equation (10.1.1.5) into (10.1.1.4),

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{4\pi^2}{\lambda^2} \psi = 0 \quad (10.1.1.6)$$

we know,

$$\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \nabla^2$$

We can write equation (10.1.1.6) as,

$$\nabla^2 \psi + \frac{4\pi^2}{\lambda^2} \psi = 0 \quad (10.1.1.7)$$

Also by de Broglie relation, we have $\lambda = \frac{h}{p}$,

Equation (10.1.1.7) becomes,

$$\nabla^2 \psi + \frac{p^2}{\hbar^2} \psi = 0 \quad \text{with } [\hbar = \frac{h}{2\pi}] \quad (10.1.1.8)$$

Total energy of the particle,

$$E = K + V$$

$$\text{i.e. } K = E - V$$

$$\frac{1}{2}mv^2 = E - V$$

$$m^2v^2 = p^2 = 2m(E - V)$$

Substituting in equation (10.1.1.8) to get,

$$\nabla^2 \psi + \frac{2m(E - V)}{\hbar^2} \psi = 0 \quad (10.1.1.9)$$

This is the time-independent Schrödinger wave equation.

For a free particle, $V = 0$, the equation becomes,

$$\nabla^2 \psi + \frac{2mE}{\hbar^2} \psi = 0 \quad (10.1.1.10)$$

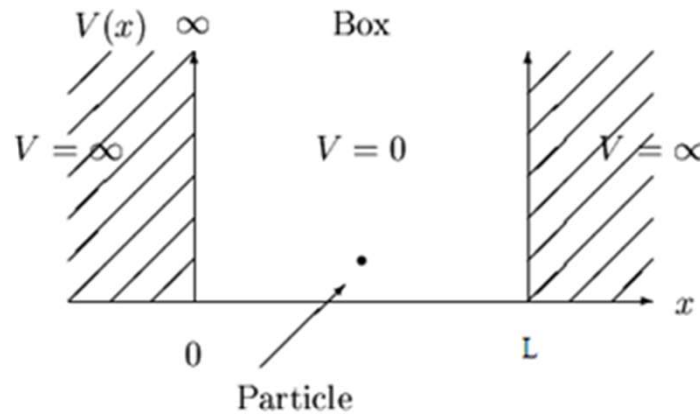
For one-dimensional case (x-axis only), the equation becomes,

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2mE}{\hbar^2} \psi = 0 \quad (10.1.1.11)$$

The solutions of this equation are called steady-state wave functions.

The obtained energy values are called **Eigen values** and corresponding wave-functions are called **Eigen functions**.

Particle in an one-dimensional infinitely deep potential well



The particle motion is restricted to x-axis between $x=0$ to $x=L$.

$$\psi = 0 \quad \text{for } x \leq 0 \text{ and for } x \geq L$$

$$\psi = 0 \quad \text{for } 0 \geq x \geq L$$

$$\text{within the box } \psi = \psi \text{ and } V(x) = 0 \quad \text{for } 0 < x < L$$

Thus within the box the Schrodinger equation will be

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} E\psi = 0 \quad (10.2.1.1) \quad (\text{Since } V(x) = 0)$$

Solution of this equation will be,

$$\psi = A \sin kx + B \cos kx \quad (10.2.1.2)$$

$$\text{where } k^2 = \frac{2m}{\hbar^2} E \quad \text{or} \quad k = \frac{\sqrt{2mE}}{\hbar}$$

The boundary conditions are

$$\psi = 0 \quad \text{at} \quad x = 0 \quad \text{and} \quad \psi = 0 \quad \text{at} \quad x = L$$

For first boundary condition

$$0 = A * 0 + B * 1 \quad \Rightarrow \quad B = 0$$

$$\text{So that} \quad \psi = A \sin kx \quad (10.2.1.3)$$

Now from 2nd boundary condition at $x = L$, $\psi = 0$

$$0 = A \sin kL$$

Since $A \neq 0$ This is only possible when

$$\sin kL = 0 \quad \text{or}$$

$$kL = n\pi \quad (10.2.1.4) \quad \text{where } n = 1, 2, 3, 4, 5, \dots$$

let us substitute the value of k in equation (10.2.1.4)

$$\frac{\sqrt{2mE}}{\hbar} L = n\pi$$

$$\frac{2mE_n}{\hbar^2} L^2 = n^2 \pi^2 \quad (\text{One can replace } E \text{ to } E_n \text{ because energy is changing with } n)$$

or

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2} \quad (10.2.1.5) \quad \text{where } n = 1, 2, 3, 4, \dots$$

Thus energy of the particle can have only the values given by equation (10.2.1.5) . These values are the eigen values of energy.

$$\psi_n = A \sin \frac{\sqrt{2mE_n}}{\hbar} x \quad (10.2.1.6)$$

Substituting for E_n

$$\psi_n = A \sin \left(\frac{\sqrt{2m}}{\hbar} \frac{n\pi\hbar}{L\sqrt{2m}} x \right)$$

$$\psi_n = A \sin \frac{n\pi x}{L} \quad (10.2.1.7)$$

ψ_n is called as the eigen function corresponding to eigen value E_n .

The requirement is that the Schrödinger state equation can be solved to get the value E_n , which are called as an energy eigen values and corresponding functions are called as the eigen functions. For each quantum number n, ψ_n is a single valued function of x. Also

$\frac{\partial \psi_n}{\partial x}$ is continuous.

$$\int_{-\infty}^{\infty} |\psi_n|^2 dx = \int_0^L |\psi_n|^2 dx \quad \text{Because in other region } \psi = 0$$

$$\begin{aligned} &= A^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = \frac{A^2}{2} \int_0^L \left(1 - \cos \frac{2n\pi x}{L} \right) dx \\ &= \frac{A^2}{2} \left[\int_0^L dx - \int_0^L \cos \frac{2n\pi x}{L} dx \right] = \frac{A^2}{2} [L - 0] = \frac{A^2}{2} L \end{aligned}$$

To normalize ψ $\int_{-\infty}^{\infty} |\psi_n|^2 dx = 1$

This is because of the fact that the particle is certainly somewhere in the box, so the probability of finding it must be 1. It means that the probability density integral over all space must yield the value 1. Therefore,

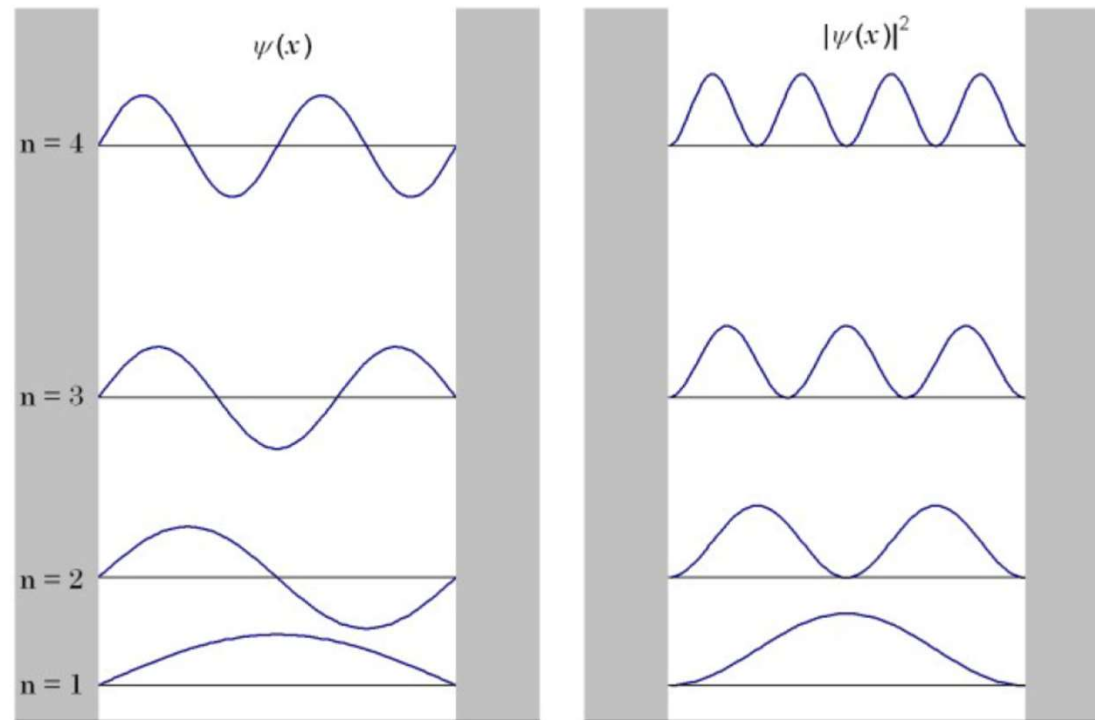
$$\frac{A^2}{2} L = 1$$

$$\text{or } A = \frac{\sqrt{2}}{L}$$

Thus the normalized wave function for the particle in the box is

$$\psi_n = \frac{\sqrt{2}}{L} \sin \frac{n\pi x}{L}$$

The normalized wave functions and probabilities are shown below.



First we notice that the wave function $\psi_1, \psi_2, \dots$ etc represents the vibration similar to string with node at two ends and antinodes in between. In $|\psi_1|^2$ the particle is most likely to be in the middle, whereas in $|\psi_2|^2$ it never exists in the middle.